Jude Rifai

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Education

Georgia Institute of Technology — Atlanta, GA
BS in Mechanical Engineering; Aerospace Engineering/Robotics Minors; GPA: 4.0/4.0 — *Expected December 2027*
Relevant coursework: Thermodynamics, Fluid Mechanics, CAD, Dynamics, Deformable Bodies

Experience

Engine Development Engineering at Yellow Jacket Space Program — Jan. 2025 – Present
Georgia Institute of Technology — *Atlanta, GA*

- Led a 2-member subteam in the design and analysis of turbopump housing components, leveraging FEA and CFTurbo to inform the volute architecture
- Directed transmission component analysis and presented spline vs. keyway trade-offs to the propulsion team, resulting in an 80% reduction in local shear stress
- Owned the design and implementation of two new fittings and a fuel tank as the Responsible Engineer, overseeing fabrication, integration, and system validation, reducing fuel fill time by 75%
- Conducted transient heat transfer analysis and cryogenic boil-off measurements to optimize LOX loading procedures, improving accuracy in consumption estimates and minimizing waste by 8%

Hypersonics Engineering Research Assistant — Jan. 2025 – Dec. 2025
Georgia Institute of Technology — *Atlanta, GA*

- Spearheaded a plasma jet experiment, coordinating instrumentation integration and data acquisition to evaluate Pressure and Temperature Sensitive Paints, improving accuracy by 10%
- Developed surface temperature and pressure probes for the team's sonic fatigue facility, enabling critical measurements under extreme conditions of 190 dB and 200°C
- Authored 3 sections on measuring article deformation, crack propagation, and internal stresses within a Hypersonic environment, documenting the instrumentation and methodology for the team's upcoming Hypersonics textbook

United Road Sustainability Advising Internship — Nov. 2023 – Jan. 2024
United Road Services — *Chattanooga, TN*

- Assessed 10 emerging zero-emission engines for United Road's car-hauling fleet, identifying the most cost-effective engines by analyzing weight, durability, and total ownership cost
- Researched regulatory frameworks across 13 states and federal legislation to ensure compliance with evolving environmental standards, narrowing feasible engines to FCEV and EV technologies
- Advised executive leadership to adopt EVs over FCEVs, citing 24% projected battery performance improvement and charging infrastructure, influencing long-term fleet electrification strategy

UTC Biomechatronics Engineering Internship — Jan. 2023 – Apr. 2023
University of Tennessee at Chattanooga — *Chattanooga, TN*

- Prototyped a Linearly Actuated Arm Brace using SolidWorks and 3D Printing, iterating to result with an elbow joint, wrist strap, and linear actuators
- Performed stress tolerance tests and evaluations on the design based on industry standards, resulting in a 5% increase in the arm brace's load-bearing capacity
- Authored and disseminated a detailed research report at the UTC Technology Symposium to an audience of 50+ scientists and professors that encapsulated design choices and manufacturing insights

Skills

Computing: SolidWorks, CFTurbo, C#, C++, Python, Java, MATLAB, Arduino, JavaScript, React, NodeJS, FEA
Manufacturing: Waterjet Cutting, Laser Cutting, Machining (Mill, Lathe), 3D Printing
Languages: English (native), Spanish (fluent), Arabic (fluent)
Clubs/Involvement: GT Climbing Club, Investments Committee, Startup Exchange
Interests: Karate, Climbing (V5), Jiu-Jitsu, Classical Guitar, Electric Guitar, Piano